Inventory Spreadsheet Data — Problems & Resolution Log

Project: WizKlub Inventory bootstrap
Date: September 2026
Source files: `FINAL_DATA.xlsx`, `school_kits_distribution.xlsx`, `SC_WIZKLUB_Combined_Single_Sheet.xlsx`

This document records every significant data problem discovered while preparing inventory spreadsheets for MongoDB seeding, why each problem occurred, and how it was resolved (or what remains open).

Clean outputs live in the user's `Downloads/final data/` folder. Regenerate with:

```
node backend/inventory/scripts/buildCleanInventorySpreadsheets.js
```

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1. Executive summary

Area	Problem count	Status
Startup crash (duplicate SC codes)	2 critical	Fixed in clean data (4 schools excluded)
FINAL_DATA duplicate / alias rows	14 codes	Fixed in clean data
Wrong sheet for pending schools	41 + 36 schools	Fixed — split by ack status
Cross-sheet overlaps	15 + 2 schools	Fixed — merged into combined
SC code conflicts (Gurugram / Valasaravakkam)	1 pair	Fixed — codes reassigned
Schools missing from all clean sheets	4 schools	**Open — needs new SC codes**
Flat kit counts in distribution	41 schools	Noted — combined sheet preferred for /26

Clean output files: `FINAL_DATA_CLEAN_164.xlsx`, `SC_WIZKLUB_Combined_CLEAN_36.xlsx`, `school_kits_distribution_CLEAN.xlsx` (empty)

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2. Problem: Docker startup crash (E11000 duplicate key)

Symptom

Backend container crash-looped on startup with:

```
E11000 duplicate key error collection: wizklub-inventory.schools
index: code_1 dup key: { code: "SC-024/24" }
```

Root cause

`syncFinalDataSpreadsheet.js` reads **every row** in `FINAL_DATA.xlsx` and calls `School.create()` when no match is found. The MongoDB `School` model requires **unique** `code` (SC code). Two rows shared the same code:

SC Code	School 1	School 2
SC-024/24	Kompally 2 (196 kits)	Mehdipatnam (181 kits)
SC-028/22	International school Vijayawada (239 kits)	Thoraipakkam (329 kits)

Where it failed

```
start-backends.sh
  → ensureExcelInventorySync.js
    → repairExcelInventoryTotals.js
      → syncFinalDataSpreadsheet.js  →  School.create()  💥
```

Resolution

- Removed both rows for each broken code from acknowledged data (4 schools, 945 kits excluded until fixed).
- Clean FINAL_DATA has **no duplicate SC codes**.

Still open

Assign **new unique SC codes** to Mehdipatnam and Thoraipakkam (and confirm Vijayawada vs Thoraipakkam split), then re-add to FINAL_DATA.

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3. Problem: FINAL_DATA mixed acknowledged and pending schools

Symptom

`FINAL_DATA.xlsx` had **197 rows** but only **156** had kit data. **41 rows** had empty / null kits.

Root cause

FINAL_DATA was treated as "all schools" instead of **"sent and acknowledged only"**. The 41 no-kit schools were actually **pending acknowledgment** — their kit data lives in `school_kits_distribution.xlsx`.

Resolution

- **FINAL_DATA** = only rows **with kits** (acknowledged).
- **41 no-kit schools** → sourced from `school_kits_distribution` or `SC_WIZKLUB_Combined` as pending.

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4. Problem: 14 duplicate SC codes in FINAL_DATA

Symptom

14 SC codes appeared **twice** in FINAL_DATA (28 rows).

Two categories

A. Alias duplicates (12 codes) — naming variants, one empty

Same SC code, two spellings; one row had kits, one did not.

SC Code	Empty row (pending)	Row with kits (ack)
SC-002/22	Coimbatore	coimbatore 1
SC-002/23	Kaggadsapura 3	Kaggadasapura 3
SC-004/25	Nashik	Nashik 1
SC-006/23	Aurangabad	AURANGABAD 1
SC-009/23	Kaggadasapura Sanskriti	Kaggadasapura 2
SC-010/24	Varthur	Future Pathways Global School Varthur
SC-021/22	Hosur	Hosur 1
SC-025/22	Tumkur	Tumkur 1
SC-034/24	Ghaziabad	Ghaziabad 1
SC-040/22	Marthahally	marthahally 1
SC-065/22	Ram Murthy Nagar	Ram Murthy Nagar 1
SC-078/22	New Town	New Town 1

Risk: Importing both rows creates duplicate schools or double-counted kits.

Resolution: Keep only the **with-kits** row in FINAL_DATA. Empty alias → pending in distribution (or already merged).

B. Broken duplicates (2 codes) — see Section 2

Both rows had kits; different real schools shared one code.

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5. Problem: Three sheets with unclear roles

Symptom

Same schools appeared in multiple files with different kit counts and statuses.

Intended model (after cleanup)

Sheet	Meaning	School count	Kit total
FINAL_DATA	Sent **and acknowledged**	164	41,726
SC_WIZKLUB_Combined	2026 joined, sent, **not** ack	36 (24 SC + 12 NSC)	9,517
school_kits_distribution	Was pending ack (older)	0 (merged)	—

Root cause

Startup script loaded only `FINAL_DATA.xlsx` + old split not-ack files (`SC-SCHOOL-NOT_ACKNOWLEDGED.xlsx` , `WIZKLUB_NOT_ACKNOWLEDGED.xlsx`), not the user's updated distribution / combined sheets.

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6. Problem: 15 SC `/26` schools in both distribution and combined

Symptom

After normalizing combined codes to `/26`, **15 schools** appeared in **both** `school_kits_distribution` and `SC_WIZKLUB_Combined`.

Root cause

2026 schools were copied into distribution (flat 293 kits each) while also existing in the combined sheet with **detailed per-kit rows**.

Risk

Double-loading pending dispatches for the same school.

Resolution

- Remove all 15 from distribution.
- Use **combined sheet only** for those schools.

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7. Problem: Gurugram 12 vs Valasaravakkam SC code conflict

Symptom

Same SC code used for **different schools** across files:

Sheet	SC-012/26	SC-013/26
school_kits_distribution	Valasaravakkam	Gurugram 12
SC_WIZKLUB_Combined	Gurugram 12	—

Resolution (agreed mapping)

School	Correct code	Source sheet
Gurugram 12	**SC-012/26**	Combined
Valasaravakkam	**SC-013/26**	Combined (moved from distribution)

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8. Problem: 24 distribution schools “move to acknowledged” — only 12 were new

Symptom

Plan was: move all **24** non-`/26` schools from distribution to FINAL as acknowledged → expect **176** schools (152 + 24).

Root cause

12 of the 24 shared SC codes with schools **already in FINAL_DATA** under different names (alias pairs from Section 4A).

Resolution

- **12 new schools** added to FINAL (Agra, Bikaner, Dehradun, etc.).
- **12 alias schools** marked "already acknowledged" — not duplicated.
- Final count: **164 schools**, not 176.

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9. Problem: Flat 293 kit count in distribution

Symptom

Almost every school in `school_kits_distribution.xlsx` showed exactly **293 kits**.

Root cause

Wide-format placeholder export, not actual per-school totals.

Impact

Combined sheet has accurate per-kit breakdown for 2026 schools. Agra 2 and Valasaravakkam still use 293-kit wide data from distribution.

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10. Problem: Four schools excluded from all clean data

School	SC Code	Kits	Issue
Kompally 2	SC-024/24	196	Shares code with Mehdipatnam
Mehdipatnam	SC-024/24	181	Shares code with Kompally 2
International school Vijayawada	SC-028/22	239	Shares code with Thoraipakkam
Thoraipakkam	SC-028/22	329	Different region from Vijayawada

Total excluded kits: 945 — **action required** from operations.

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11. Infrastructure issues (related)

Issue	Workaround
`host.docker.internal` fails on Windows host	Run scripts inside Docker container or use `127.0.0.1`
Partial seed + restart loop	Drop DB before re-seed after data fixes

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12. Code / startup gaps

Gap	Detail
Startup loads all FINAL rows	Should skip empty-kit rows

No dedup by SC code	Should keep one row per code
Old not-ack files in Docker image	vs user's combined sheet
`school_kits_distribution` not in startup	Script exists but not wired in `start-backends.sh`

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13. Recommended next steps

1. Assign new SC codes for Mehdipatnam and Thoraipakkam.
2. Copy clean xlsx files into `backend/inventory/data/` .
3. Update startup scripts to use clean files and skip empty/duplicate rows.
4. Drop DB and rebuild containers.

See also: `INVENTORY_DATA_AUDIT_REPORT.md` *in Downloads,* `scripts/buildCleanInventorySpreadsheets.js`